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Algorithms going wild – A review of machine learning techniques for terrestrial ecology

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ABSTRACT

The integration of artificial intelligence (AI) algorithms in ecological research is revolutionizing how we monitor, predict, and manage natural systems, enabling more advanced data analysis, pattern recognition, and predictive modelling. This review critically analyzes and synthesizes the application of machine learning and deep learning in terrestrial ecology, providing a comprehensive overview of their paradigms – namely unsupervised, supervised, and reinforcement learning – and semi-supervised learning, along with their respective algorithm families, strengths, and limitations. We examine both current and emerging applications in terrestrial ecological dynamics and modelling, ecosystem management and conservation, identification and classification tasks, such as trait and behavior recognition. Despite these advancements, we summarize several issues hindering the extensive adoption of AI algorithms in ecology, such as inconsistencies or limitations in datasets, algorithm complexity and interpretability affecting transparency and reliability, high computational demands raising environmental sustainability concerns, and difficulties with model generalization. To address these barriers, we identify key areas for future research, namely optimizing data collection, using transfer learning and data augmentation, refining model transparency through explainable AI (XAI) and ethical considerations, and integrating causal inference into AI models. We conclude that AI algorithms hold great promise for delivering more accurate, scalable, and timely data, advancing real-time monitoring and near-instantaneous predictions – e.g., seasonal forecasting – for more dynamic responses to environmental changes. The need for continued methodological innovation and multi- and trans-disciplinary collaboration is emphasized to ensure these technologies are effective, sustainable, and equitable in supporting ecosystem conservation and restoration efforts addressing global ecological crises.

Abbreviations

AI	Artificial Intelligence
BART	Bayesian Additive Regression Tree
BRT	Boosted Regression Tree
CNN	Convolutional Neural Network
DBSCAN	Density-Based Spatial Clustering Of Applications with Noise
DL	Deep Learning
DNN	Deep Neural Network
ExT	Extra Trees
GAN	Generative Adversarial Network
GBM	Gradient Boosting Machine
GCN	Graph Convolutional Network
GPU	Graphic Processing Unit
L1	L1 Regularization – LASSO (Least Absolute Shrinkage and Selection Operator) Regression
L2	L2 Regularization – Ridge Regression
ML	Machine Learning
NN	Neural Network
OTU	Operational Taxonomic Unit
PCA	Principal Component Analysis
RF	Random Forest
RL	Reinforcement Learning
RNN	Recurrent Neural Network
RoF	Rotation Forest
RS	Random Subspace method
SGB	Stochastic Gradient Boosting
SOM	Self-Organizing Map

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SVM	Support Vector Machine
t-SNE	t-Distributed Stochastic Neighbor Embedding
VAE	Variational Autoencoder
XAI	Explainable Artificial Intelligence
XGB	Extreme Gradient Boosting

Introduction to terms

In this review, we make a distinction between two uses of the term machine learning (ML). When we use ML alone, we are referring specifically to traditional ML or classical ML methods – simpler, traditional algorithms such as decision trees, linear regression, and support vector machines, excluding deep learning (DL) and reinforcement learning (RL). For the broader category that includes classical ML, DL, and RL, we refer to it as general ML. This distinction is used consistently throughout the text to clarify which level of complexity is being discussed.

Moreover, while reinforcement learning is one of the three main paradigms in machine learning, it is discussed separately throughout this review due to its unique application to dynamic, decision-making environments, which differ significantly from more traditional machine learning approaches that rely on static datasets.

1. Introduction

In recent years, the field of terrestrial ecology has witnessed a transformative shift in its approach to data analysis and modelling, owing largely to the advent of machine learning (ML), deep learning (DL), and partially to reinforcement learning (RL) techniques (Jordan and Mitchell, 2015; Thessen, 2016; Christin et al., 2019; Lapeyrolerie et al., 2022). General ML, a subset of artificial intelligence (AI), refers to the development of algorithms that enable computers to learn from and make predictions or decisions based on data, without being explicitly programmed (Zhou, 2021). DL, a more advanced form of ML, uses artificial neural networks with multiple layers to automatically extract complex features from data (LeCun, Bengio, and Hinton, 2015). In ML, three different main paradigms are employed to address specific types of tasks, each drawing on distinct approaches to learning. Supervised learning relies on examples with known outcomes to guide the learning process, while unsupervised learning aims to uncover underlying structures or patterns within a dataset without predefined outcomes. Reinforcement learning (RL) takes a unique approach by focusing on learning through interaction with an environment, optimizing model performance through trial and error (Sutton and Barto, 2018). While RL shares foundational concepts with the other ML paradigms, it is often discussed separately due to its specialized focus on adaptive decision-making and optimization, making it a distinct approach compared to more traditional ML methods, where data is typically static. Additionally, semi-supervised learning combines aspects of supervised and unsupervised approaches to improve accuracy but does not constitute a separate paradigm. Both ML and DL offer ecologists new opportunities to analyze vast amounts of ecological data, uncover hidden patterns, and make accurate predictions across various spatial and temporal scales (Pichler and Hartig, 2023; Borowiec et al., 2022). RL, on the other hand, provides a framework for decision-making and optimization in dynamic ecological systems (Lapeyrolerie et al., 2022). These computational methodologies have indeed emerged as powerful tools for understanding complex ecological systems, predicting ecological patterns and processes, and informing management and conservation strategies (Pichler and Hartig, 2023). As ecologists cope with increasingly large and diverse datasets, the application of general ML methods holds promise for unlocking new insights into terrestrial ecological dynamics and addressing pressing environmental challenges (Reichstein et al., 2019).

1.1. Historical overview

The groundwork for ML emerged in the 1950s, especially with Alan Turing's seminal question, *Can machines think?* which he addressed in his 1950 work *Computing Machinery and Intelligence*, where he also introduced the concept of the Turing Test to assess a machine's intelligence (Turing, 1950). Around this time, Arthur Samuel at IBM developed a checkers-playing program that improved over time by learning from past games, marking one of the earliest examples of a program with a learning component, and he is also credited with coining the term machine learning in 1959 (Chandra and Hareendran, 2021). Early ML development (1960s-1980s) was heavily influenced by statistical methods and pattern recognition. For instance, Frank Rosenblatt's Perceptron (Rosenblatt, 1958) – one of the first artificial neural networks capable of binary classification – inspired further research on neural networks, even though its limitations (Minsky and Papert, 1969) initially slowed the field's progress. In the 1970s, interest in Bayesian inference and probabilistic modeling grew, as did the development of fundamental ML algorithms such as k-nearest neighbors and decision trees (Bansal et al., 2022), which are still widely used and will be discussed later on in this review.

The 1980s brought renewed focus on neural networks with the development of the backpropagation algorithm enabling effective training of multi-layer networks, marking a period of algorithmic and computational advancements extending into the 1990s (Leung and Haykin, 1991). The rise of other essential algorithms, such as support vector machines (SVMs) in the early 1990s, provided a solid method for classification tasks (Hearst et al., 1998). The first comprehensive work synthesizing scientific applications of various ML approaches in ecology dates back to 1999 and was authored by Alan Fielding (Fielding, 1999). Concurrently, advances in computing power enabled researchers to process larger datasets, critical for training more complex models (Pichler and Hartig, 2023).

Perry et al. (2022) identify the mid-2000s as the onset of an explosion in publications on AI applications within the environmental sciences, marking it as the beginning of exponential growth in scientific literature on the topic. The advent of the internet led to massive amounts of data available for training ML models and popularized ensemble methods like boosting and bagging algorithms, which began to challenge traditional statistical models.

In the past decade, the accessibility of graphics processing units (GPUs) facilitated the rapid spread of DL, and thus the practical implementation of deep neural networks (DNNs), characterized by multiple hidden layers allowing breakthroughs in image (Krizhevsky, Sutskever, and Hinton, 2017) and speech recognition (Hinton et al., 2012) tasks. Key milestones, such as the development of convolutional neural networks (CNNs) and recurrent neural networks (RNNs), revolutionized the field and paved the way for applications across diverse domains, including ecology (O'Shea and Nash, 2015; Salehinejad et al., 2017).

1.2. Contemporary overview in terrestrial ecology

In contemporary terrestrial ecology, general ML algorithms are employed across several research areas to tackle complex ecological challenges. These techniques are used to analyze and interpret large-scale ecological datasets, ranging from satellite imagery or more in general remote sensing data (Zhang et al., 2016; Zhu et al., 2017) to species occurrence records and environmental variables (Liu et al., 2018; Beery et al., 2021). In species distribution modelling, for example, ML models are used to project in space and time species' habitat suitability and distribution patterns based on environmental covariates (Beery et al., 2021; Zurell et al., 2020; Thuiller et al., 2009; Elith and Leathwick, 2009). On the other hand, DL approaches, such as CNNs, have demonstrated remarkable performance in image-based ecological tasks such as habitat classification and species monitoring using camera trap data (Nguyen et al., 2017; Norouzzadeh et al., 2018; Willi et al.,

2018; Schneider et al., 2019). Additionally, RL algorithms are used to optimize management strategies in dynamic ecosystems, such as forest fire management and invasive species control (Taleghan et al., 2015; Subramanian and Crowley, 2018; Frankenhuis et al., 2019; Antonelli et al., 2022). General ML algorithms are helpful in ecological forecasting, ecosystem monitoring, and conservation planning, enabling ecologists to make more informed decisions and address pressing environmental issues in a rapidly changing world (Pichler and Hartig, 2023). Given the proliferation of diverse tools within the field of AI, confusion in the application of its various techniques also increased (Pilon et al., 2023). This challenge is compounded by the rapid pace of AI and ML advancements, which frequently outstrip the capacity of ecologists and practitioners to fully grasp or keep up with evolving methodologies, terminology, and best practices, thus creating possible implementation barriers – especially in ecological contexts where specialized ML expertise may be limited. Bridging this knowledge gap makes it critical to distill and elucidate the foremost applications of general ML in terrestrial ecology, aimed at enlightening key directions for interdisciplinary exploration and advancing ecological understanding. Such efforts will require interdisciplinary collaboration, clear documentation, and accessible training resources tailored specifically for ecological researchers.

1.3. Objective of the review

The purpose of this review is to provide a comprehensive overview of the applications, challenges, and current directions of ML, DL, and RL in terrestrial ecology. By synthesizing existing scientific literature, this review aims to (a) examine the diverse applications of general ML techniques in terrestrial ecological research; (b) identify key challenges and bottlenecks hindering the widespread adoption of AI algorithms in ecology in general, such as data availability, algorithm complexity, and interpretability; (c) explore potential avenues for overcoming these challenges and advancing the integration of AI into ecological research and practice, highlighting emerging developments and promising research directions in the field.

Through this review, we seek to provide ecologists, researchers, and conservation practitioners with a comprehensive understanding of the potential of ML, DL, and RL in terrestrial ecology, and to catalyze further interdisciplinary collaborations and innovations in this rapidly evolving field.

2. Review method

On May 1st, 2024, we conducted an inquiry using the Web of Science database, through the following string of keywords encompassing diverse facets of ecological research and artificial intelligence methodologies.

ALL=(("ecolog*" OR "ecolog* model*" OR "biodiversity" OR "species distribution model*" OR "remote sensing") AND ("AI" OR "artificial intelligence" OR "machine learning" OR "deep learning" OR "supervised learning" OR "unsupervised learning" OR "semi-supervised learning" OR "reinforcement learning"))

Subsequent refinement of the search parameters involved limiting the search to articles published between 2015 and 2024. This yielded a corpus of 1679 unique papers for further scrutiny. Following this initial identification phase, the full-text articles underwent screening and evaluation to ascertain their eligibility for inclusion, with emphasis placed on the relevance of content and the extent of citations garnered. As our initial keyword search was skewed towards modelling and biodiversity, we soon found this approach to be overly restrictive. Additional sources identified outside this search were more pertinent and offered opportunities to broaden the scope of our review. Moreover, the lack of refinement in our search string allowed the keywords to appear anywhere within the publications, resulting in a less targeted set of initial findings. This limitation also explains why "terrestrial ecology" was not initially included as a keyword; while our investigation ultimately focused on terrestrial ecosystems, the initial search aimed to capture a broad spectrum of AI applications in ecology without preemptively constraining the scope. However, as the review progressed, our refined selection criteria ensured a targeted emphasis on terrestrial systems, aligning with the study's objectives.

Therefore, in refining the scope of our analysis, papers judged to be extraneous to the scope of our investigation were discarded. We excluded studies on freshwater ecosystems to maintain a concentrated emphasis on terrestrial ecology, which we define as the study of ecological processes, interactions, and patterns on ecosystems mostly based on land dynamics and relying on related resources, including species dynamics, ecosystem functions, and environmental responses. Agricultural ecosystems were not excluded a priori, but we chose not to put as much emphasis on them, as there is already extensive literature on AI applications in agricultural settings (Liakos et al., 2018; Jha et al., 2019; Benos et al., 2021; Sharma et al., 2020; Saleem et al., 2021; Eli-Chukwu, 2019). Our focus was instead on natural and semi-natural terrestrial environments where human management is present but does not primarily aim at maximizing production. While various managed ecosystems, such as planted forests and enclosed grasslands, were considered, agricultural studies, which often center on yield optimization and intensive resource management, were not the primary emphasis of our review. Similarly, aquatic ecosystems, particularly freshwater environments, were excluded due to their distinct hydrological and ecological dynamics. These criteria were set to better define our study's objective.

As additional papers were incorporated subsequent to a thorough examination of publications citing them, the methodological approach culminated in the utilization of 326 papers for the ensuing analysis. An overarching trend gleaned from the reviewed literature was the temporal distribution of publications, with a discernible surge in interest evident in works disseminated in recent years, indicative of an evolving landscape of inquiry within the field. Due to nuanced boundaries between algorithm categories, the organization of the following paragraphs is neither exhaustive nor entirely precise.

3. General ML algorithms

The central concept behind ML algorithms involves taking a particular algorithmic structure and adjusting its parameters based on the provided data during training. This process occurs while concurrently optimizing the model's complexity to strike a balance between bias and variance, aiming for effective generalization to unseen data. ML encompasses a diverse range of algorithmic structures trained to perform specific tasks by minimizing a general loss function, not necessarily linked to the probability of the observations (Shmueli, 2010). Tasks serve as the practical applications of these algorithms, and they can be broadly categorized into several types, each with distinct characteristics and applications (Fig. 1, Fig. 2).

ML and DL algorithms can be trained using datasets that contain labeled data in supervised learning, unlabeled data in unsupervised learning, or a combination of both in semi-supervised learning (Berry et al., 2020; Van Engelen and Hoos, 2019). Labeled data consists of input-output pairs, where each input is associated with a corresponding output label, whereas unlabeled data lacks explicit output labels.

3.1. Unsupervised learning algorithms

Unsupervised learning has strong foundations in neuroscience, influenced by pioneers like Horace Barlow for neural coding, Donald MacKay's cybernetic approach, and David Marr's learning models of the neocortex. The Hebb rule (Hebb, 1949) linked statistical methods to neurophysiological plasticity, setting a foundation for future learning models. Geoffrey Hinton and Terrence Sejnowski further advanced the field with the Boltzmann machine (Hinton and Sejnowski, 1986),

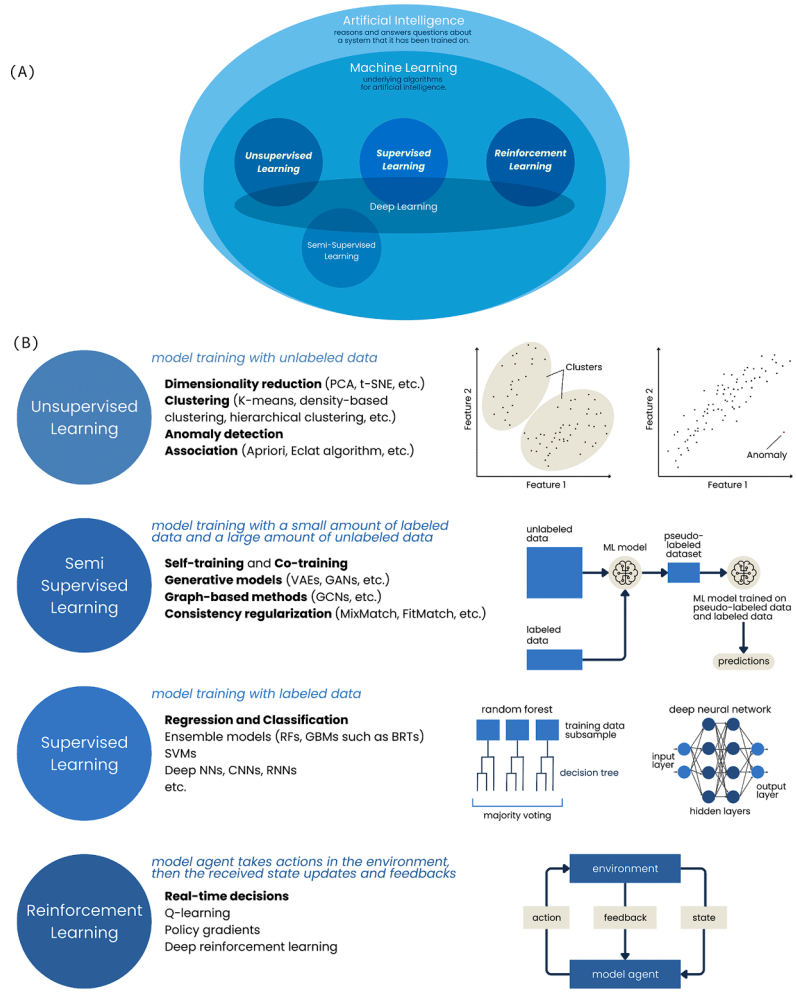

Fig. 1. (A) Venn diagram illustrating the relationships between AI, general ML, ML paradigms (supervised learning, unsupervised learning, and reinforcement learning – shown in bold and italics), semi-supervised learning methods, and DL. (B) An overview of learning approaches in general ML, with corresponding visual examples for each.

bringing statistical principles to density estimation. Despite these advancements, feature extraction methods have historically received less focus in the development of unsupervised learning techniques (Dayan et al., 1999).

Algorithms used in unsupervised learning in machine learning examine and spot patterns in data without the requirement for labeled outputs; thus, the algorithm must locate hidden structures or patterns in the data without direct supervision (Krotov and Hopfield, 2019). Unsupervised learning algorithms contribute to a greater knowledge of complex data relationships and enable smart data-driven decision--making processes by revealing hidden patterns and structures inside datasets (Berry et al., 2020). Dimensionality reduction, clustering, and anomaly detection – where the model lowers the complexity of the data, finds groups, or finds outliers in the data – are important uses of unsupervised learning.

In ecology, unsupervised learning is particularly useful when dealing with complex, high-dimensional datasets such as those involving genetic, geographic, environmental, and phenotypic information. For example, methods like self-organizing maps (SOMs) can learn associations between multiple input features to identify distinct clusters, which may correspond to species or other ecological entities (Chon, 2011), therefore enabling species delimitation and the detection of meaningful ecological groupings based on diverse data layers. Unsupervised learning also helps integrate different types of ecological data, ensuring that no single layer (e.g., genetic data) dominates the analysis, allowing for a more comprehensive understanding of species boundaries and conservation needs (Pyron, 2023).

3.1.1. *Dimensionality-reduction*

Unsupervised learning algorithms like principal component analysis (PCA) and t-distributed stochastic neighbor embedding (t-SNE) aim to reduce the number of features in a dataset while preserving its essential information, aiding in visualizing high-dimensional data and identifying underlying structures (Jolliffe and Cadima, 2016; Hinton and Roweis, 2002). In particular, PCA reduces dimensionality by projecting data onto orthogonal components that maximize variance (Jolliffe and Cadima, 2016), t-SNE reduces high-dimensional data to two or three dimensions for visualization while preserving local structure (Hinton and Roweis, 2002). Lastly, another promising approach are autoencoders, which are neural networks (see 3.2.3) used for dimensionality reduction and feature learning by encoding and decoding data (Bank et al., 2023).

3.1.2. *Clustering*

On the other hand, clustering algorithms, such as K-means clustering (Hartigan and Wong, 1979; Forgy, 1965), hierarchical clustering (Nielsen, 2016), density-based clustering – e.g., density-based spatial clustering of applications with noise, or DBSCAN (Ester et al., 1996), and unsupervised random forest (Mantero and Ishwaran, 2021), group data points based on similarity metrics, enabling the identification of natural

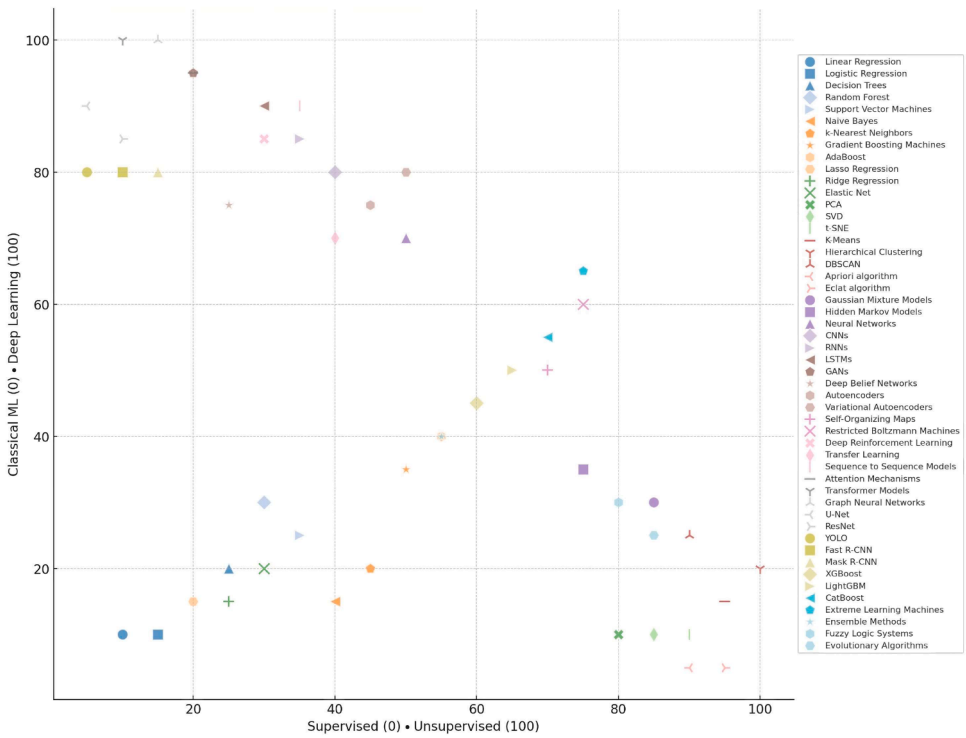

Fig. 2. The scatter plot categorizes various ML and DL algorithms, ranging from 0 (fully supervised) to 100 (fully unsupervised) on the x-axis and from 0 (classical ML) to 100 (DL) on the y-axis. Supervised learning algorithms are those that require labeled data to learn, whereas unsupervised algorithms do not. Classical ML algorithms are typically simpler models, whereas DL involves complex structures. The plot visually represents where these algorithms stand in terms of their learning style and complexity. The legend on the right side of the plot matches each algorithm with its corresponding symbol and color. The categorization was based on a structured evaluation considering the following criteria for the supervised to unsupervised axis (x-axis) and classical ML to DL axis (y-axis). For the x-axis, methods that rely predominantly on labeled data (e.g., logistic regression, decision trees) are placed towards the Supervised (left) end, methods that are traditionally used for clustering or pattern discovery without labeled data (e.g., k-means, DBSCAN) are placed towards the Unsupervised (right) end, whereas some methods, such as neural networks and autoencoders, can be applied in both settings and are thus placed in intermediate positions. For the y-axis, classical machine learning techniques (e.g., linear regression, support vector machines) are positioned towards the Classical ML (bottom) end, methods relying on deep architectures with multiple layers (e.g., CNNs, GANs, transformers) are positioned towards the Deep Learning (top) end, whereas hybrid models or algorithms that incorporate deep learning principles but maintain classical elements (e.g., gradient boosting, transfer learning) are placed in intermediate positions. To ensure consistency, these placements are based on commonly accepted categorizations in literature, empirical usage patterns, and expert judgment.

clusters within the dataset. Clustering is useful for segmenting data into meaningful groups without prior knowledge of class labels. Specifically, K-means partitions data into k clusters by minimizing within-cluster variance (Forgy, 1965), hierarchical clustering builds a hierarchy of clusters using agglomerative (bottom-up) or divisive (top-down) approaches (Nielsen, 2016), and DBSCAN identifies clusters of varying shapes based on density and can handle noise/outliers (Ester et al., 1996).

3.1.3. Anomaly detection

Lastly, unsupervised learning algorithms can detect outliers or anomalies in data that deviate significantly from the norm (Rehman and Belhaouari, 2021). Techniques like isolation forest (Liu et al., 2008) and one-class SVM (Shin et al., 2005) are commonly used for anomaly detection, with isolation forest detecting anomalies by isolating observations in a feature space (Liu et al., 2008), and one-class SVM – a variant of SVMs (see 3.2.2) – identifying outliers in a dataset (Shin et al., 2005).

3.2. Supervised learning algorithms

Supervised learning in general ML involves training algorithms on labeled data to make predictions or decisions based on input-output pairs, meaning that a labeled dataset with input feature labels and corresponding output labels is needed for supervised learning algorithms to function. The algorithm learns from these examples to identify patterns and relationships in the data (Lu, 2010). During the training phase, the algorithm adjusts its parameters iteratively to minimize a predefined loss function, which quantifies the disparity between predicted outputs and actual labels (Mohri et al., 2018). Once the model is trained, it can make predictions on new, unseen data by applying the learned patterns to input features and generating output predictions. The model's performance is then evaluated using metrics such as accuracy – which measures the proportion of correct predictions out of all predictions made, precision – measuring the proportion of true positive predictions among all positive predictions made by the model, recall or sensitivity – which measures the proportion of true positive predictions among all actual positives, and F1 score – the harmonic mean of precision and recall, balancing the trade-off between the two (Grandini et al., 2020). Support vector machines (SVMs), ensemble models, and neural networks (NNs) are common algorithms for supervised learning, each with unique characteristics and applications, as well as strengths and limitations. The choice of algorithm depends on the specific task, the nature of the data, and the desired outcome (Pichler and Hartig, 2023).

In their early conception, supervised learning algorithms were seen as tools for addressing regression or classification problems, based on the relationship between a set of explanatory variables and a defined target variable. Since 1995, a group of methods has gained prominence among the numerous available approaches due to their non-parametric nature and independence from distributional assumptions. These methods were highly versatile, with extensions that enable the modelling of multidimensional or functional outputs (Crisci et al., 2012). For

instance, nowadays supervised learning can be used in ecology to predict biotic index values, which assess environmental impact, from eDNA metabarcoding data without depending on the taxonomic identification of sequences. Models are trained on data from various ribosomal bacterial and eukaryotic markers to evaluate their effectiveness in detecting ecological disturbances. Results show accurate predictive models with any molecular marker that encompasses a broad range of potential bioindicator taxa, outperforming traditional taxonomy-based methods and demonstrating how supervised learning can improve biodiversity monitoring through the use of molecular data to assess environmental changes (Cordier et al., 2018).

3.2.1. Ensemble models (bagging, boosting, stacking)

Ensemble models combine multiple base learners – i.e., individual models that serve as building blocks in ensemble methods – to improve predictive performance and generalization (Schapire, 1989; Friedman, 2001). Bagging methods, such as random forest (RF) and extra trees (ExT), use bootstrap aggregating, which involves creating multiple subsets of the training data through bootstrapping (random sampling with replacement) and aggregating their predictions, thus they build multiple decision trees and aggregate their predictions through voting or averaging, increasing model strength and reducing variance (Breiman, 2001; Abbasi et al., 2022). Instead, boosting methods like gradient boosting machines (GBMs), extreme gradient boosting (XGB), stochastic gradient boosting (SGB), Bayesian additive regression trees (BARTs), and AdaBoost sequentially build weak learners – i.e., models that perform slightly better than random guessing, used to iteratively improve performance – to correct errors made by previous models, leading to strong predictive capabilities (Friedman, 2001; Abbasi et al., 2022). Their prominence in terrestrial ecological modelling is emphasized by boosted regression trees (BRTs) for species distribution models (Elith et al., 2008; Thuiller et al., 2009; Elith and Leathwick, 2009). Additionally, the random subspace method (RS), rotation forest (RoF), and stacked-based models further extend the ensemble approach by enriching diversity and improving accuracy (Abbasi et al., 2022). The augmentation of simplistic algorithms or statistical models into more sophisticated ensembles presents a pathway towards greater complexity, albeit at the expense of interpretability, while frequently yielding diminished prediction errors. BRTs and RFs persist as commonly used methodologies in ecology (e.g., beyond species distribution models, for linking gene variation to phenotypes or predictions in ecological networks) since they are particularly valued for their adeptness in handling smaller datasets when contrasted with analogous DL models, hence their ability to balance predictive accuracy with data availability and computational efficiency (Elith and Leathwick, 2009; Brieuc et al., 2018; Pichler et al., 2019).

3.2.2. Support vector machines

SVM is a powerful algorithm that functions as a binary classifier, distinguishing between two classes by establishing an optimal hyperplane within the feature space. This hyperplane serves as a boundary, effectively partitioning the feature space to segregate data points belonging to different classes (Brereton and Lloyd, 2010). While originally designed for binary classification tasks, SVMs can be extended to handle multiclass classification and regression tasks through various techniques (Hsu and Lin, 2002). In essence, SVM identifies the most suitable hyperplane that maximizes the margin, or the distance between the hyperplane and the nearest data points from each class, and by doing so SVM not only separates the classes but also seeks to achieve the widest possible gap between them. This strategy improves the model's strength and generalization capability, making it less prone to errors on unseen data (Brereton and Lloyd, 2010). SVMs are not limited to linear separability as they can effectively handle nonlinear relationships between features and classes by employing a kernel function, which maps the original feature space into a higher-dimensional space where linear separation is feasible (Cortes and Vapnik, 1995). Consequently, SVMs can tackle complex classification problems even when the underlying relationships between features and classes are nonlinear (Pigozzi and Genesereth, 2022). Before the advent of DL, SVMs were the prevailing technique for image classification in ecological studies, notably in the domain of remote sensing (Melgani and Bruzzone, 2004; Pal and Mather, 2005).

3.2.3. Neural networks

NNs are a class of algorithms inspired by the human brain's neural structure, consisting of interconnected layers of nodes called neurons (Rosenblatt, 1958). Each neuron receives input signals, performs a computation on them, and then passes the result to the neurons in the next layer. Between the input and output layers are intermediate layers called hidden layers, which perform complex computations on the input data through a series of weighted connections and nonlinear activation functions (Bishop, 2016). NNs learn complex patterns and hierarchies in large datasets through forward and backward propagation, adjusting weights to minimize prediction errors. The number of hidden layers and neurons within each layer can vary depending on the complexity of the problem (Bishop, 2016). Already in December 1998, the first workshop on the applications of this approach in ecological modeling was organized in Toulouse, France, where numerous works were presented on topics such as remote sensing data, population and community ecology, ecosystems, and global change and ecosystem sensitivity (Lek and Guégan, 1999). Lecun et al. (2015) describe how the discussion has evolved, expanding from traditional neural networks to more advanced DNN techniques.

DNNs, including convolutional neural networks (CNNs) and recurrent neural networks (RNNs), are characterized by multiple hidden layers and perform tasks like image recognition, acoustic processing, and time series analysis (Venkatesan and Li, 2017; Hewamalage et al., 2021). Specifically, CNNs are widely used for image recognition and classification tasks through convolutional layers that help extract spatial hierarchies of features from input images, or more precisely from grid-like data (Venkatesan and Li, 2017). On the other hand, RNNs are designed to handle sequential data by maintaining internal states as well as processing inputs in a sequential manner, and they are commonly used in tasks such as time series prediction and acoustic processing (Hewamalage et al., 2021).

While commonly applied in supervised learning tasks, NNs are highly versatile and can be applied across different learning paradigms as well as semi-supervised learning frameworks depending on the task requirements, making such networks valuable for a wide range of ecological and analytical applications.

3.3. Semi-supervised learning algorithms

Semi-supervised learning falls between supervised and unsupervised learning, with the main idea behind it being to improve model performance by using both labeled and unlabeled datasets for training the algorithm. Semi-supervised learning algorithms have the potential to achieve better generalization than supervised learning algorithms trained exclusively on labeled data because they use the unlabeled data to capture additional information about the underlying data distribution and the labeled data to guide the learning process (Zhu and Goldberg, 2009). Combining supervised and unsupervised learning strategies is a popular approach in semi-supervised learning (Maturo and Verde, 2022; Van Engelen and Hoos, 2019; Kostopoulos et al., 2018). For instance, in a semi-supervised approach the labeled data may be used to train an initial model and the unlabeled data could be used to fine-tune the model's parameters or learn more representations of the data. Another strategy involves transferring information from the labeled to the unlabeled data by creating a related surrogate task or proxy objective, which uses the patterns learned from the labeled data to provide guidance for learning on the unlabeled data (Zhai et al., 2019; Chapelle et al., 2010). For example, this approach has been used in terrestrial ecology to

improve species classification performance by leveraging the natural variations found in image collections from static monitoring cameras. Instead of just altering images or guessing which images are different, extra context data – like the time and place when each picture was taken – was used to find pairs of images that are likely to show the same subject (Pantazis et al., 2021). In situations when labeled data is scarce or expensive to obtain, but unlabeled data is abundant, semi-supervised learning algorithms come in handy. However, to properly use the information at hand, selecting the right learning objectives and giving careful thought to the link and balance between labeled and unlabeled data (Maturo and Verde, 2022) are necessary for designing effective semi-supervised learning algorithms (Van Engelen and Hoos, 2019).

For example, self-supervised learning can be used to analyze remote sensing imagery by creating pretext tasks, which are designed to generate pseudo-labels from large amounts of unlabeled data, reducing the need for manual annotation. This semi-supervised learning approach enables the mapping of forest disturbances, such as windthrow and snow load damage, using data from satellite imagery (Kuzu et al., 2024).

3.3.1. Self-training and co-training

In semi-supervised learning, self-training algorithms start with a small labeled dataset to train an initial model, which is then used to iteratively label the unlabeled data. This process involves retraining the model with the newly labeled data, gradually expanding the dataset with pseudo-labeled examples (Chapelle et al., 2010). Co-training, on the other hand, involves training two classifiers on different views or subsets of the data (e.g., different feature sets). Each classifier is used to label the unlabeled data, with the results from one classifier helping to train the other, thereby iteratively refining the model's accuracy (Blum and Mitchell, 1998).

3.3.2. Generative models

Generative models are also common in semi-supervised learning, including algorithms like variational autoencoders (VAEs) and generative adversarial networks (GANs). VAEs combine generative modelling with semi-supervised learning by simultaneously generating data and classifying it, providing a way to use both labeled and unlabeled data (Kingma and Welling, 2013). GANs, on the other hand, involve training a generative model to produce realistic data samples and a discriminator to differentiate between real and generated samples. The discriminator's feedback is used to refine the labeling process of the unlabeled data, improving the model's overall performance (Goodfellow et al., 2020).

3.3.3. Graph-based methods

Graph-based methods are another important category in semi-supervised learning, with algorithms such as label propagation and graph convolutional networks (GCNs). Label propagation spreads labels from labeled to unlabeled data points across a graph based on their similarity, effectively using the structure of the data to inform classification (Zhu and Ghahramani, 2002). Instead, GCNs extend traditional convolutional neural networks (see 3.2.3) to work with graph-structured data, making them particularly useful for classification tasks where data points have complex, interdependent relationships (Scarselli et al., 2009).

3.3.4. Consistency regularization

Lastly, consistency regularization is a strategy that ensures model predictions remain stable under variations like data augmentation or noise (Englesson and Azizpour, 2021). This approach is commonly used in methods like MixMatch (D. Berthelot et al., 2019) or FixMatch (Sohn et al., 2020), where the goal is to enforce consistency between the model's predictions on altered versions of the same data, thereby improving its generalization to unseen data.

3.4. Deep learning

The DL approach arises from the necessity to efficiently represent complex functions that shallow NNs struggle to model without requiring exponentially more neurons. As highlighted by (Wang and Raj, 2017) in their work on the origins of DL, deep architectures provide a more practical solution to this challenge. Bengio and Delalleau (2011) argued that deep networks draw inspiration from the hierarchical structure of the human brain, where abstract concepts are built from simpler ones. Studies by Hastad (1986) and Delalleau and Bengio further demonstrated that deep networks can represent functions with polynomially many neurons, whereas shallow networks require exponentially more resources. Eldan and Shamir (Eldan and Shamir, 2016) solidified these findings, showing that depth offers exponential advantages over width for most activation functions. Despite this theoretical strength, deep networks face practical difficulties, particularly in parameter optimization, which is addressed through techniques like backpropagation.

DL shares foundational concepts with other ML algorithms, however it is often regarded as a distinct domain due to its unique principles and task-specific architectures (Box 1), diverging from traditional ML approaches (Bengio, 2009). The level of sophistication and intricacy within the architectural design of a model profoundly impacts the model's capacity to discern patterns within datasets and subsequently make accurate predictions. In traditional ML models, such as decision trees and SVMs, model complexity traditionally hinges upon factors such as the number of features, the choice of algorithms, and hyperparameters like regularization strength (Yang and Shami, 2020). Conversely, DL models are inherently complex (Box 2) due to a multitude of contributing factors, stemming from the depth and width of these networks. Foremost among these is the architectural depth, typified by the presence of multiple hidden layers between the input and output layers (Bengio, 2009). In DL, the hidden layers are determined by the model architecture, which is designed by the user, but their parameters (weights and biases) are learned and optimized by the algorithm during training. This architectural characteristic affords DL models the capacity to discern complex patterns within the data, as it facilitates the acquisition of hierarchical representations of underlying features (LeCun, Bengio, and Hinton, 2015). The number of neurons in each layer of a DL model determines its capacity to learn and represent features from the input data, therefore increasing the number of neurons can improve the model's ability to capture complex patterns without overfitting, at least for in-distribution predictions; extrapolation beyond the data domain remains challenging for ML as for other approaches. (Pichler and Hartig, 2023). Surprisingly, increasing the depth and width beyond a certain size reduces generalization error, indicating larger model sizes can decrease total prediction error, contrary to expectations (Arora et al., 2019; Huh et al., 2021; Nakkiran et al., 2021; Novak et al., 2020; Shwartz-Ziv and Alemi, 2019).

DL is used in ecology to automate tasks such as species identification (Ruff et al., 2021; Torney et al., 2019) and environmental monitoring (Yuan et al., 2020) by recognizing complex patterns in data, including images, sounds, and molecular sequences. Most studies employ CNNs for supervised learning tasks like image recognition and classification, but DL is also applied to molecular data (Lamperti et al., 2023) and ecological modelling (Borowiec et al., 2022). It offers an alternative to traditional mechanistic models by handling complex, high-dimensional data with strong predictive performance, although it is sensitive to bias in input data. The use of DL in ecology is rapidly expanding, particularly in biodiversity monitoring and data-driven discovery (Borowiec et al., 2022).

3.5. Reinforcement learning algorithms

As with many other approaches in general ML, RL finds its historical

***Box 1* – Complexity differences between ML and DL models**

While both ML and DL models involve managing model complexity to achieve optimal performance, DL models are characterized by their deep architectures, automatic feature learning capabilities, and ability to handle complex relationships in data. Understanding the differences in model complexity between ML and DL is essential for selecting the appropriate approach for different ecological research tasks and applications.

Depth and breadth – DL models are typically deeper and wider than traditional ML models, with multiple hidden layers and a large number of parameters. This increased depth and breadth enable DL models to learn complex representations and abstract features from data.

Feature engineering – This is the process of selecting, transforming, and creating input variables. In ML, feature engineering plays a crucial role in designing informative features for the model. In contrast, DL models can automatically learn relevant features from raw data, reducing manual feature engineering.

Interpretability – Traditional ML models are often more interpretable than DL models, which are sometimes referred to as "black boxes" due to their complex architectures. Understanding and interpreting DL model decisions can be challenging.

Computational resources – DL models require significant computational resources for training and inference due to their complex architectures and large datasets. ML models, especially simpler ones, may be more computationally efficient.

Data size – Due to their complexity and numerous parameters, DL models typically require large amounts of data to train effectively, whereas some ML models can perform well with smaller datasets, as these models have simpler structures.

***Box 2* – Adjusting model complexity**

In order to strike a balance between model performance, interpretability, computational efficiency, and task requirements in general ML applications, adjusting model complexity is needed. The overarching goal in ML is to compromise between model bias, which may result in underfitting (the model lacks the capacity to capture patterns in the data), and variance, which may manifest as overfitting (the model learns noise in the data instead of underlying patterns), by adjusting the complexity of the model to attain optimal performance across both training and test datasets (Pichler and Hartig, 2023). Model complexity is typically managed through optimization techniques aimed at mitigating the risk of overfitting, hence fine-tuning models for better predictive accuracy, while simultaneously enhancing the model's ability to generalize well to unseen data (Yang and Shami, 2020). The choice of model complexity should align with the specific requirements of the task at hand. For some tasks, simpler models may suffice and offer advantages such as easier implementation, faster training, and better explainability (Pichler and Hartig, 2023). Conversely, deeper and wider neural networks could actually be beneficial for reducing total prediction error (Arora et al., 2019; Huh et al., 2021). Internal and external optimization are two key approaches used in ML to adjust model complexity and improve performance.

Internal complexity optimization – Internal, or algorithmic, complexity optimization is the process within ML algorithms that enables them to adjust their model complexity autonomously. Several ML algorithms use ensemble predictions, either implicitly or explicitly, by combining multiple models to generate a final prediction. In ensemble models the effective complexity is not merely the sum of the individual complexities of each member, instead it is influenced by the average complexity of the ensemble members and the variations between them. This diversity among ensemble members helps mitigate errors and variance in the final prediction (Bernardo and Smith, 2009; Dormann et al., 2018; Ganaie et al., 2022). To facilitate internal complexity optimization, ML algorithms frequently include mechanisms aimed at increasing heterogeneity within the ensemble, for instance through bagging (Sagi and Rokach, 2018), RF (Breiman, 2001), and gradient boosting (Friedman, 2001), thereby improving the overall performance and strength of the model.

External complexity optimization – On the other hand, external complexity optimization involves an additional step of fine-tuning the hyperparameters of a model, which are settings that define the model's architecture and how it learns from data (Pichler and Hartig, 2023). In practice, hyperparameters are optimized through techniques like cross-validation, where the model is trained and evaluated on different subsets of the data to find the optimal hyperparameter values, helping prevent overfitting and ensuring better generalization to unseen data (Stone, 1974). Different ML algorithms have varying hyperparameters to optimize; for example, RF models have fewer hyperparameters, while DL models require extensive tuning of their network architecture and training parameters. To manage the risk of overfitting and control model complexity, various regularization methods are commonly used – including dropout, L1 (Tibshirani, 1996) and L2 (Hoerl and Kennard, 1970) regularization, and batch normalization (Ioffe and Szegedy, 2015). More specifically, dropout randomly ignores some neurons during training to reduce dependency on specific neurons, improving generalization. L1 regularization (Tibshirani, 1996), or Lasso, adds a penalty based on the absolute values of the model's weights, encouraging the model to set less important feature weights to zero, which effectively performs feature selection. Conversely, L2 regularization (Hoerl and Kennard, 1970), or Ridge, adds a penalty based on the squared values of the weights, shrinking the weights but not necessarily reducing them to zero, helping to keep all features in play while controlling their influence. Lastly, batch normalization (Ioffe and Szegedy, 2015) normalizes the input to each layer of a neural network, which stabilizes the training process and can speed up learning. Overall, these methods enhance the model's ability to generalize by reducing its reliance on patterns that are specific to the training data.

roots in psychological theorizing. According to Sutton and Barto (2018), RL represents a fusion of the trial-and-error "law-of-effect" framework from psychology, optimal control theory from engineering, the secondary reinforcement tradition in learning, and the use of decaying stimulus traces. Examples of the latter include Hull's concept of a goal gradient (Hull, 1952) and Wagner's model of conditioning (Wagner, 1981).

RL is a general ML paradigm focused on decision-making and learning optimal behavior through environmental interaction. In RL, an agent learns to achieve a goal by taking actions in an environment and receiving immediate feedback in the form of rewards or penalties, with the goal being to maximize the cumulative reward it receives over time.

At each time step, the agent observes the environment's state and it selects an action based on its policy, then the agent transitions to a new state after taking an action, and receives a reward signal that indicates the immediate desirability of the action. Ultimately, the agent's objective is to learn a policy that maximizes long-term cumulative rewards (Kaelbling et al., 1996). One of the key features of RL is the exploration-exploitation trade-off, where the agent must balance between exploiting its current knowledge to maximize immediate rewards and exploring new actions or states to discover better long-term strategies. Notably, the agent relies on its own experiences and exploration rather than access to an exact model of the environment, in an approach that resembles active learning, with the agent actively seeking new information to boost performance (Van Otterlo and Wiering, 2012). RL algorithms, such as Q-learning (Li, 2023), policy gradients (Zhang et al., 2020), and deep reinforcement learning (Li, 2023), have been developed to address various challenges in learning from interaction and have demonstrated their capabilities in learning complex behaviors from raw sensory inputs (Chapman et al., 2023; Subramanian and Crowley, 2018).

In ecological contexts, RL can be used to model how organisms or agents learn and adapt in realistic and dynamic environments without resets, where feedback is infrequent or delayed. Unlike simpler, episodic models, non-episodic RL reflects the ongoing interactions organisms have with their environment, where they must learn behaviors without clear or immediate rewards. Understanding environmental properties like how much the environment changes independently of an organism's actions can help better predict how species adapt to varying conditions over time, making it easier for RL models to simulate ecological behaviors by providing a more natural context for learning (Co-Reyes et al., 2020).

4. Applications in terrestrial ecology

The application of different ML methods in terrestrial ecology is essential to address the complexity and heterogeneity of ecological systems, as ecosystem processes are inherently dynamic, nonlinear, and influenced by spatial and temporal variability, therefore necessitating diverse analytical approaches. For instance, supervised learning techniques are widely used for predictive modelling, such as species distribution modelling (Beery et al., 2021; Elith et al., 2008; Elith and Leathwick, 2009) and land cover classification (M et al., 2023), whereas unsupervised learning is helpful with pattern recognition in large ecological datasets, including habitat clustering and anomaly detection. On the other hand, reinforcement learning can support decision-making frameworks in conservation and resource management, where adaptive strategies are required. Therefore, the selection of an appropriate ML approach depends on the research objective, data structure, and the specific ecological processes under investigation, highlighting the need for methodological diversity in terrestrial ecology.

In recent years the role of general ML in terrestrial ecology has expanded, marking a shift in ecological research and practice. AI, particularly through ML methods, has become increasingly present in ecological applications, offering powerful tools for data analysis, modelling, and prediction tasks (Almeida et al., 2024; Ryo, 2023; Olden et al., 2008). While general ML techniques have diversified to progressively encompass DL in more recent years, ML methods continue to prevail in the landscape of ecological applications (Strydom et al., 2021). Simplified ML algorithms, such as RF and GBM, remain widely used due to their effectiveness, interpretability, and computational efficiency (Abbasi et al., 2022). Despite the emergence of more complex AI approaches, the enduring popularity of ML methods stresses their continued relevance and practicality in addressing ecological challenges across various spatial and temporal scales. As ecological datasets continue to grow in size and complexity, the role of ML and DL is expected to further expand (Pichler and Hartig, 2023; Borowiec et al., 2022), driving improvements in the field and advancing our understanding of ecological systems.

4.1. Ecological dynamics and modelling

Ensemble models such as RF and boosting algorithms are commonly used in species distribution modelling (Beery et al., 2021; Elith et al., 2008; Elith and Leathwick, 2009) and have shown effectiveness in predicting interactions within ecological networks as well, since these models can capture complex relationships between species or components in ecological systems, hence providing further comprehension of network dynamics (Dallas et al., 2021; Pichler et al., 2019; Desjardins-Proulx et al., 2017). Ensemble models are used to predict various ecological phenomena, ranging from species responses to environmental changes (Hao et al., 2020, 2019; Grenouillet et al., 2011; Thuiller et al., 2009), disease outbreaks (Becker et al., 2022; Wardeh et al., 2021; Ray and Reich, 2018), invasive species spread (Barbet-Massin et al., 2018; Gallien et al., 2010; Stohlgren et al., 2010), to broader ecosystem dynamics such as the study of complex ecological networks, including food webs (Desjardins-Proulx et al., 2017) and mutualistic interactions (Pichler et al., 2019). Additionally, in community ecology and biodiversity studies, DL algorithms can help understand community structure, species interactions, biodiversity patterns, and ecological community dynamics, ultimately contributing to a deeper knowledge of ecosystem functioning and resilience (Christin et al., 2019; Chen et al., 2016; Desjardins-Proulx et al., 2017).

4.2. Ecosystem management and conservation

Ecologists need to monitor and understand ecosystem changes due to human activities for effective management and conservation (Ellis, 2015). AI techniques are commonly used tools that allow practitioners to improve their decision-making processes, optimize resource utilization, and promote sustainable practices for preserving ecosystems. ML and DL tools aid in conservation planning and management efforts as they identify key habitats and restoration areas (Cheng et al., 2018; Duhart et al., 2018; Moradi et al., 2019), help assess invasive species risk (Jensen et al., 2020; Barbet-Massin et al., 2018; Gallien et al., 2010; Stohlgren et al., 2010), and are also employed for biodiversity assessments (Distler et al., 2015), forest management (Lauer et al., 2019), ecosystem service management (Scowen et al., 2021), and wildlife management (Humphries et al., 2018). Although agricultural ecosystems are strongly influenced by human activities and their primary goal is not strictly conservation-oriented, they are considered an integral part of terrestrial ecosystems, as these environments involve crucial ecological processes and interact with surrounding natural and semi-natural ecosystems. The application of AI and ML in agriculture provides useful insights into ecological dynamics, resource optimization, and sustainable land management (Ye et al., 2024), yet, while including agricultural ecosystems is important when discussing managed contexts in terrestrial ecology, the extensive body of literature on this topic warrants a dedicated review, and therefore, given the focus of our study, a comprehensive assessment of agricultural applications is beyond its scope.

Conversely, RL, with its capacity for solving sequential decision problems through trial and error with feedback from the environment (Li, 2023), allows adaptive approaches that continually refine decisions based on measured impacts within ecosystems (Lapeyrolerie et al., 2022). RL is particularly well-suited for adaptive management practices; for instance, learning optimal interventions like controlled burns (Viseras et al., 2024; Hammond et al., 2020). Furthermore, RL facilitates wildlife protection efforts by learning patrol routes that optimally detect and deter poaching based on patterns of past attacks and monitoring (Tambe and Parkes, 2022), continuously improving enforcement efficiency. RL can also aid in natural resource allocation, demonstrated through a forest management scenario where agents balance economic and ecological goals amid varying disturbances (Bone and Dragićević,

***Box 3* – Main ML and DL algorithms**

Linear Regression – A supervised algorithm used for predicting a continuous outcome variable based on one or more predictor variables (Maulud and Abdulazeez, 2020).

Logistic Regression – A statistical model that in its basic form uses a logistic function to model a binary dependent variable. In regression analysis, logistic regression is estimating the parameters of a logistic model; it is a form of binomial regression (LaValley, 2008).

Decision Trees – Supervised algorithms that model decisions and their possible consequences, resembling a tree structure (Kotsiantis, 2011).

Random Forest – An ensemble of decision trees, typically used for classification and regression tasks (Biau and Scornet, 2016).

Support Vector Machines – Supervised algorithms that find the hyperplane that best separates classes of data in feature space (Hearst et al., 1998).

Naive Bayes – A set of supervised learning algorithms based on applying Bayes' theorem with the "naive" assumption of independence between every pair of features (Rish, 2001).

k-Nearest Neighbors – A simple algorithm that stores all available cases and classifies new cases based on a similarity measure, e.g., distance functions (Laaksonen and Oja, 2002).

Gradient Boosting Machines – An ensemble technique that builds models sequentially, with each new model attempting to correct the errors of the previous ones (Natekin and Knoll, 2013).

AdaBoost (Adaptive Boosting) – A ML meta-algorithm that can be used in conjunction with many other types of learning algorithms to improve performance (Cao et al., 2013).

LASSO Regression (Least Absolute Shrinkage and Selection Operator) – A type of linear regression that uses shrinkage. Shrinkage is where data values are shrunk towards a central point, like the mean (Kukreja et al., 2006).

Ridge Regression – Also known as Tikhonov regularization, this is a method of estimating the coefficients of multiple-regression models in scenarios where independent variables are highly correlated (Johansen, 1997).

Elastic Net – A regularized regression method that linearly combines the L1 and L2 penalties of LASSO and ridge methods (De Mol et al., 2009).

PCA (Principal Component Analysis) – A statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables (Jolliffe and Cadima, 2016).

SVD (Singular Value Decomposition) – A method used in linear algebra to decompose a matrix into three constituent matrices, uncovering the intrinsic geometric structure of the matrix (Klema and Laub, 1980).

t-SNE (t-Distributed Stochastic Neighbor Embedding) – A ML algorithm for visualization. It is a nonlinear dimensionality reduction technique well-suited for embedding high-dimensional data for visualization in a low-dimensional space of two or three dimensions (Hinton and Roweis, 2002).

K-Means – A type of unsupervised learning, used when you have unlabeled data (i.e., data without defined categories or groups). The goal of this algorithm is to find groups in the data, with the number of groups represented by the variable K (Hartigan and Wong, 1979).

Hierarchical Clustering – A method of cluster analysis which seeks to build a hierarchy of clusters. Strategies for hierarchical clustering generally fall into two types: agglomerative and divisive (Nielsen, 2016).

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) – A data clustering algorithm that is commonly used in data mining and ML. Based on a set of points, DBSCAN groups together points that are closely packed together, marking as outliers points that lie alone in low-density regions (Schubert et al., 2017).

Apriori algorithm – An algorithm for frequent item set mining and association rule learning over relational databases. It proceeds by identifying the frequent individual items in the database and extending them to larger and larger item sets as long as those item sets appear sufficiently often in the database (Hegland, 2007).

Eclat algorithm – A depth-first search algorithm used in frequent item set mining and association rule learning. It is similar to the Apriori algorithm but uses a different data structure to store the candidate item sets (Borgelt, 2003).

Gaussian Mixture Models – A probabilistic model that assumes all data points are generated from a mixture of a finite number of Gaussian distributions with unknown parameters (Reynolds, 2009).

Hidden Markov Models – A statistical Markov model in which the system being modeled is assumed to be a Markov process with unobservable (hidden) states (Rabiner and Juang, 1986).

Neural Networks – A network or circuit of neurons or, in a modern sense, an artificial NN, composed of artificial neurons or nodes. These are a series of algorithms that mimic the operations of a human brain to recognize relationships between vast amounts of data (Gurney, 2018).

CNNs (Convolutional Neural Networks) – DL algorithms that are particularly powerful for analyzing visual imagery (Gu et al., 2017).

RNNs (Recurrent Neural Networks) – NNs designed for handling sequential data, with the ability to use their internal memory to process inputs over time, making them ideal for tasks like language processing and speech recognition (Salem, 2022).

LSTMs (Long Short-Term Memory networks) – A type of RNN that is capable of learning order dependence in sequence prediction problems (Malhotra et al., 2015).

GANs (Generative Adversarial Networks) – An approach to generative modelling using DL methods, such as CNNs, that are trained with two models: one to generate candidates and one to evaluate them (I. Goodfellow et al., 2020).

Deep Belief Networks – A generative graphical model, or alternatively a class of DNN, composed of multiple layers of latent variables (hidden units) with connections between the layers but not between units within each layer (Hinton, 2009).

Autoencoders – A type of artificial NN used to learn efficient codings of unlabeled data (unsupervised learning). It learns two functions: an encoding function that transforms the input data, and a decoding function that recreates the input data from the encoded representation (Bank et al., 2023).

Variational Autoencoders – An artificial NN architecture that combines probabilistic graphical models and variational Bayesian methods. It maps each input into a distribution within the latent space and then decodes from the latent space to the input space according to a distribution (Doersch, 2016).

Self-Organizing Maps – An unsupervised ML technique used to produce a low-dimensional (typically two-dimensional) representation of a higher-dimensional data set while preserving the topological structure of the data (Kohonen, 1990, 2002).

Restricted Boltzmann Machines – A generative stochastic artificial NN that can learn a probability distribution over its set of inputs. It is a variant of Boltzmann machines, with the restriction that their neurons must form a bipartite graph (Fischer and Igel, 2012).

Deep Reinforcement Learning – A subfield of ML that combines RL and DL. RL considers the problem of a computational agent learning to make decisions by trial and error, and deep reinforcement learning incorporates DL into the solution (Arulkumaran et al., 2017).

Transfer Learning – The reuse of a pre-trained model on a new problem. It allows a machine to exploit the knowledge gained from a previous task to improve generalization about another (Weiss et al., 2016).

Sequence to Sequence Models – Models used in sequence prediction tasks, such as language modelling and machine translation. They use one LSTM as the encoder to read the input sequence and another LSTM as the decoder to extract the output sequence from the context vector (Chiu et al., 2018).

Attention Mechanisms – ML-based mechanisms that calculate "soft" weights for each word, more precisely for its embedding, in the context window. These weights can be computed either in parallel or sequentially (Niu et al., 2021).

Transformer Models – DL models introduced in 2017, fundamental in natural language processing, and applied to a wide range of tasks in ML and artificial intelligence. They are based on the multi-head attention mechanism (Amatriain, 2023).

Graph Neural Networks – Generally, they are NNs that directly operate on the graph structure. They take a graph as input and return a graph as output, where node features and structure are encoded within the network (Wu et al., 2020; Zhou et al., 2020).

U-Net – A CNN architecture developed for image segmentation. It is designed to work with fewer training images and yield more precise segmentation (Siddique et al., 2021).

ResNet (Residual Neural Network) – A type of DL model that uses residual functions with reference to the layer inputs, allowing the training of very deep networks (Xie et al., 2016).

YOLO (You Only Look Once) – An object detection model known for its speed and accuracy, using a single NN to detect objects in real-time (Redmon et al., 2016).

Fast R-CNN – An object detection model that improves on the previous R-CNN by using a region proposal network to efficiently classify object proposals using deep convolutional networks (Jiang and Learned-Miller, 2017).

Mask R-CNN – An extension of Fast R-CNN that adds a branch for predicting segmentation masks for each instance, enabling precise pixel-wise instance segmentation (He et al., 2018).

XGBoost (Extreme Gradient Boosting) – A scalable and distributed gradient-boosted decision tree ML library, used for regression, classification, and ranking problems (Chen et al., 2014).

LightGBM (Light Gradient Boosting Machine) – A gradient-boosting framework that uses decision tree algorithms and is designed for performance and scalability (Ke et al., 2017).

CatBoost – A high-performance library for gradient boosting on decision trees, effective for handling datasets with categorical features and used for classification, regression, and ranking tasks (Prokhorenkova et al., 2018).

Extreme Learning Machines – Feedforward NNs for classification, regression, clustering, and other tasks, with single or multiple layers of hidden nodes where the parameters need tuning (Huang et al., 2006).

Ensemble Methods – These methods use multiple learning algorithms to obtain better predictive performance than could be obtained from any of the constituent learning algorithms alone (Dietterich, 2000).

Fuzzy Logic Systems – A form of many-valued logic where the truth value of variables may be any real number between 0 and 1, used to handle the concept of partial truth (Mendel and Mouzouris, 1997).

Evolutionary Algorithms – A subset of evolutionary computation, these are algorithms that use mechanisms inspired by biological evolution, such as reproduction, mutation, recombination, and selection, to solve optimization problems (Vikhar, 2016).

2009), and improve forest harvesting strategies, enabling forest company agents to maximize profits while cooperating with a conservationist agent to protect habitats (Bone and Dragićević, 2010). A recent systematic review study (Araújo et al., 2023) calculated that over 36 % of Q1 articles on ML applications in agriculture from 2019 to 2023 used ensemble model techniques (RF, XGB, GBM, etc.), 25 % used techniques based on causal inference (i.e., NNs), and 16 % used supervised learning algorithms (i.e., SVMs). It has been observed that there is a significant improvement in agricultural efficiency due to ML-based precision farming techniques, particularly in resource allocation. This has led to more judicious use of inputs such as water, fertilizers, and pesticides, positively reducing the environmental footprint of agricultural practices.

4.3. Identification and classification

ML algorithms at large can classify different land cover types and monitor changes in land use over time using satellite imagery, aerial photographs, and drone data in remote sensing applications (Talukdar et al., 2020; Helber et al., 2019; Zhang et al., 2019; Abdi, 2019; Kussul et al., 2017). DL algorithms like CNNs are mostly used for image and audio analysis, making them helpful for species identification from

visual (Torney et al., 2019; Wäldchen and Mäder, 2018; Dyrmann et al., 2016; Nguyen et al., 2017; Villa et al., 2017; Lee et al., 2015) and acoustic data (Kahl et al., 2021; Ruff et al., 2021; Xu et al., 2020; LeBien et al., 2020; Ruff et al., 2019), but also from molecular and morphological data (Lamperti et al., 2023; Yang et al., 2021). Autoencoders are a promising DL-based approach gaining popularity, able to process unstructured ecological data and extract meaningful features and patterns (Pichler and Hartig, 2023); they have been used for biodiversity assessment from unlabeled audio (Rowe et al., 2021), for image-based tasks from satellite imagery (Zhou et al., 2021; Zerrouki et al., 2021), and also for genomic sequences (Wang and Gu, 2018). However, DL models often require large amounts of data to achieve high levels of accuracy in species recognition tasks, which can be challenging to obtain in ecological studies especially for rare or elusive species (McKelvey et al., 2008) or in remote habitats where data collection is limited (Harwood et al., 2016). Identification and classification undertakings can also be led by classical ML unsupervised learning algorithms. For instance, K-means groups individuals based on similarities in their characteristics, whereas t-SNE visualizes the similarities and differences between ecological samples, all of which ends up being helpful for tasks such as species delimitation – identifying potential new species or differentiating between closely related species (Derkarabetian et al., 2019) – or operational taxonomic units (OTUs) in metabarcoding (Deiner et al., 2017). Similarly, Kohonen's self-organizing maps have been used in terrestrial ecology to classify vegetation types based on environmental variables, helping to identify patterns and ecological gradients (Adamczyk et al., 2012). Unsupervised learning algorithms are also effective in detecting anomalies in ecological datasets, as these algorithms can identify unusual patterns or outliers that deviate significantly from the norm, thus highlighting potentially important ecological phenomena or errors in data collection (Tanigaki et al., 2023).

4.4. Trait and behavior recognition

DL techniques can extract complex traits and behaviors from ecological data, allowing for in-depth analysis and understanding of species characteristics such as animal movement patterns (Pereira et al., 2022; Nathan et al., 2022), plant phenology (Katal et al., 2022; Pearson et al., 2020), and animal vocalizations (Stowell, 2022), therefore supporting the study of ecological processes and behaviors (Pichler and Hartig, 2023; Borowiec et al., 2022). More specifically, DL algorithms can analyze tracking data from GPS collars or camera traps to identify animal movement patterns (Nathan et al., 2022), providing information on migration (Wijeyakulasuriya et al., 2020), habitat selection (Bergen et al., 2022), and species interactions (Pichler et al., 2019). They can also examine remote sensing data to track leaf emergence (Samiei et al., 2020), flowering times, and senescence (Katal et al., 2022), in order to better understand climate change impacts on plant communities and ecosystems. Another application of DL algorithms is the recognition of species-specific calls (Kahl et al., 2021) or communication patterns (Stowell, 2022), increasing knowledge of animal behavior, population dynamics, and community structure (Otsuka et al., 2024).

5. Challenges hindering the extensive adoption of AI in ecology

With its capacity for advanced data analysis, pattern recognition, and predictive modelling, AI has shown potential to boost ecological research and conservation endeavors (Pichler and Hartig, 2023). Nonetheless, in order to fully realize the promise of these technologies, a number of issues must be resolved. These challenges to overcome range from data availability and quality concerns to the interpretability of AI models and the requirement for interdisciplinary cooperation (Janiesch et al., 2021). Addressing such issues needs a concerted effort to improve algorithm transparency and explainability, optimize computational workflows, and promote the responsible use of AI techniques in ecological research (Pichler and Hartig, 2023). In order to improve our comprehension of complex ecological systems and to assist evidence-based decision-making for the preservation of biodiversity and the sustainability of ecosystems, it is important that we are aware of and prepared for these problems.

5.1. Data availability and quality

Ecological datasets are often heterogeneous, complex, and collected from diverse sources, leading to issues with data consistency, completeness, and standardization (Zipkin et al., 2021). The availability of large-scale, high-quality ecological data is critical for training general ML models effectively, but many ecological datasets may be limited in size or scope (Pichler and Hartig, 2023), making it challenging to develop powerful models that can generalize well across different ecosystems and scenarios (Bernardini et al., 2024). Acquiring such data may even entail significant costs or logistical hurdles, resulting in sparse or imbalanced datasets that may be hard to acquire and not fully capture the complexity of ecological systems (Poisot et al., 2019). Additionally, data quality issues such as missing values, measurement errors, and biases further complicate the use of AI in ecological studies (Alzubaidi et al., 2023; Zipkin et al., 2021), therefore significantly impacting the accuracy and reliability of AI predictions and analyses, as well as potentially leading to erroneous conclusions and ineffective decision-making in conservation and management efforts (Pichler and Hartig, 2023; Stock et al., 2023; Danovaro, 2024).

5.2. Algorithm complexity and interpretability

The complexity of AI algorithms, particularly DNNs, can hamper the understanding and interpretation of their inner workings (Dehghan-Shoar et al., 2024). The intricate architectures and large numbers of parameters characteristic of some ML and of DL algorithms result in models that operate as "black boxes" due to their inscrutable decision-making processes, making it hard to understand how they arrive at their predictions (Buhrmester et al., 2021), especially in ecological contexts where causal relationships are of interest. The process of feature learning in DL involves automatically extracting hierarchical representations from raw data, further obfuscating the relationship between input variables and model outputs (Yang and Hu, 2020). Moreover, the optimization process involved in training these models, such as stochastic gradient descent (SGD), can lead to nonlinear decision boundaries and complex interactions between features, making it difficult to interpret the learned representations (Bottou and Bousquet, 2011). Concerning RL, designing an appropriate reward function necessitates careful consideration of factors such as prediction accuracy, timeliness, relevance, and impact (Cruz and Igarashi, 2020). Evaluating the agent's performance presents additional hurdles, involving comparisons with alternative models or methodologies and assessing the long-term consequences of actions. Furthermore, the necessity for multi-agent or cooperative RL may arise due to the presence of multiple agents, interactions, and objectives within environmental systems (Canese et al., 2021). The lack of transparency in AI algorithms makes model interpretability, reproducibility, and trustworthiness consequently difficult (Von Eschenbach, 2021; Icarte et al., 2022), hindering the broader adoption of such algorithms in fields like terrestrial ecology where transparent decision-making is essential for scientific inference and environmental management (Pichler and Hartig, 2023). Optimizing these models for ecological applications involves tuning hyperparameters, selecting appropriate architectures, and addressing issues like overfitting, which are all tasks that require specialized expertise and computational resources, adding complexity to the model development process and potentially limiting the accessibility of advanced techniques to ecologists with varying levels of technical proficiency. Specifically for terrestrial ecology, in agriculture there are various types of limitations, as most farmers lack the time and digital skills to explore AI solutions independently, and for AI to be effectively integrated it must work with

existing systems and infrastructure (Javaid et al., 2022). Technology becoming more accessible could lead to semi-autonomous agriculture thanks to AI, but the reliance on limited, seasonal data and the slow development of data infrastructure make building ML models a lengthy process, therefore the use of these technologies risks remaining limited to large organizations and companies (Javaid et al., 2022).

5.3. Computational resources and environmental implications

The computational resources required for training and using complex general ML models represent a major obstacle for researchers, as these advanced models often demand substantial computing power, including high-performance computing infrastructure and specialized hardware such as GPUs (Acun et al., 2021). Training ML and DL algorithms on large ecological datasets can be computationally intensive, requiring parallel processing capabilities and efficient memory management to handle the complex calculations involved in model optimization (Thompson et al., 2020). Limited access to these computational resources can act as a bottleneck for ecologists seeking to use advanced ML techniques, given that without adequate computing resources researchers may face delays in model development, longer processing times, and constraints on the scale and complexity of analyses that can be performed (Hu et al., 2021). Computational and theoretical obstacles are well-known problems for RL methodologies as well (Fan et al., 2021), arising from the high-dimensional, nonlinear, and chaotic dynamics frequently manifested by environmental systems (Rogers et al., 2022; Meron, 2015). Collaborations with institutions or organizations that provide access to high-performance computing facilities can help overcome these issues and enable ecologists to harness the full potential of AI algorithms for advancing ecological research and conservation (Geva et al., 2020).

However, it is worth noting that the substantial computational requirements to train and operate AI models also entail significant energy implications that impact climate and environmental sustainability (Gupta et al., 2021; Thompson et al., 2020). The increase in energy demand can lead to higher carbon emissions, especially if the energy sources are non-renewable, consequently resulting in an exacerbation of climate change by contributing to the carbon footprint (Gupta et al., 2021). Moreover, the environmental toll extends beyond carbon emissions, encompassing the resource-intensive production and disposal of electronic components (Wu et al., 2021). Therefore, adopting energy-efficient algorithms (Menghani, 2023), using renewable energy sources, and implementing sustainable practices throughout the AI lifecycle should be incentivized to help mitigate such adverse effects, thus aligning technological advancement with environmental stewardship (Wu et al., 2021). Recent studies have highlighted the growing interest and importance of decarbonizing high performance computing in order to improve their environmental and financial performance (Silva et al., 2024). The debate is ongoing regarding whether the benefits of AI advancements outweigh their environmental costs.

5.4. Model generalization

Model generalization is intrinsically tied to the reliability and applicability of AI models, and when model generalization problems occur it can be due to various factors (Barbierato and Gatti, 2024; Doshi-Velez and Kim, 2018). For instance, when the training data does not fully represent the variability present in real-world ecological systems, these dataset biases lead to models that may perform well on the training data but struggle to generalize as effectively to new data (Doshi-Velez and Kim, 2018). Poor performance on unseen data can also be a consequence of overfitting, since the model learns the noise or random fluctuations in the training data rather than the underlying patterns (Ying, 2019). Lastly, the transferability of models across different ecological contexts can be tricky, as ecological systems vary in complexity, dynamics, and underlying processes, making it hard for models trained in one context to generalize well to others (Yates et al., 2018).

6. Advancing AI integration into ecology

Despite the hurdles inherent in AI methods, they still hold promise for advancing ecological research (Pichler and Hartig, 2023). Issues such as limited generalizability, interpretability, and data quality can be addressed through targeted strategies (Doshi-Velez and Kim, 2018). For instance, using transfer learning (Weiss et al., 2016) and data augmentation (Shorten and Khoshgoftaar, 2019) are both solutions to improve model performance on small datasets (Han et al., 2018), whereas explainable AI practices can enhance transparency and trustworthiness of AI models (Arrieta et al., 2020; Ehsan et al., 2021). To further reduce data-related difficulties, open science, cooperation, and the use of standardized and high-quality datasets should also be encouraged (Zizka et al., 2019; Powers and Hampton, 2018).

6.1. Optimizing data collection and collaboration

Traditional data collection methods may not always be optimized for the requirements of more advanced ML techniques, which work best when processing large volumes of complex data (Roh et al., 2021; Arora et al., 2019). Rethinking data collection with AI-based methods could help reduce labor costs and accelerate the acquisition of larger databases (Pichler and Hartig, 2023), for instance by identifying species and extracting features from raw data (Torney et al., 2019; Ruff et al., 2021; Ferreira et al., 2020). Advances in eDNA and other sensor data can also facilitate the collection and management of large datasets using AI (Lamperti et al., 2023; Pimm et al., 2015). A major advantage of machine-assisted data collection pipelines is their reusability, without the need for retraining (Pandey and Bengani, 2022). Nevertheless, few ready-to-use pipelines exist, and those available often lack performance and generalizability (Pichler and Hartig, 2023). In addition, promoting collaboration and data sharing in ecology through public datasets and platforms, as well as crowd-sourcing initiatives, betters the access to diverse and high-quality ecological data for training and validating ML and DL models (Zizka et al., 2019; Powers and Hampton, 2018). These centralized hubs can provide easy access to diverse ecological datasets, including images, audio recordings, species observations, and environmental variables. In addition, standardizing data within these repositories also ensures well-annotated, structured, and high-quality datasets for training models, with the added benefit that shared repositories promote collaboration, reproducibility, and transparency in research by facilitating data reuse and validation (Pichler and Hartig, 2023; Powers and Hampton, 2018). Through community engagement and knowledge exchange, such platforms can encourage interdisciplinary collaboration (Heberling et al., 2021) and best practices in data sharing, while also promoting open science to enhance transparency and reproducibility in ecological research (Powers and Hampton, 2018). Jointly, standardized reporting guidelines and best practices help document general ML workflows, ensuring consistency and reliability in research outcomes. In this context, international monitoring networks have never been more important and should be promoted further to support large-scale, collaborative ecological studies.

6.2. Transfer learning and data augmentation

Traditional ML algorithms may struggle to generalize effectively due to the small sample size, leading to overfitting and reduced model performance (Zhang et al., 2021). One key approach is the use of transfer learning (Weiss et al., 2016), where pre-trained models on large, diverse datasets are fine-tuned on smaller ecological datasets and, through the knowledge and features learned from the pre-trained models, it is therefore possible to improve the performance of models on specific ecological tasks with limited data (Pichler and Hartig, 2023). Transfer

learning helps transfer knowledge from one domain to another, allowing for more efficient training and better generalization on small datasets (Weiss et al., 2016). However, transfer learning is typically used for vision-based tasks and DL, not for traditional structured tabular data (Pichler and Hartig, 2023). In these types of data, the relationship between features and the response is often specific and doesn't generalize well to other tasks, which could be why DL doesn't consistently outperform traditional ML on small structured tabular datasets (Schwartz et al., 2020; Pichler et al., 2019; Arik and Pfister, 2019), a kind that is common in ecology due to the difficulty of obtaining observations and combining datasets (Strydom et al., 2021; Maglianesi et al., 2014). As a result, it's a widespread practice to compare different algorithms for a given task due to the unpredictability of trade-offs (Norberg et al., 2019; Pichler et al., 2019). In addition, data augmentation techniques can help increase the diversity and size of ecological datasets, especially in cases of limited data availability (Rondon et al., 2024; Shorten and Khoshgoftaar, 2019). Data augmentation involves artificially generating new training samples by applying transformations such as rotation, flipping, or adding noise to existing data (Christin et al., 2019). Augmenting the dataset helps provide the model with more varied examples to learn from, improving its ability to generalize and thus make accurate predictions (Nanni et al., 2020).

6.3. Explainable AI and ethics

Both interpretability and ethical considerations are critical in developing and using AI models (Arrieta et al., 2020; Jobin and Ienca, 2019). Focusing on interpretability helps with the transparency and trustworthiness of predictive models through combined methods like feature importance analysis (Mishra et al., 2021), model visualization (Arrieta et al., 2020), and post-hoc explanations (Slack et al., 2020), which are techniques that allow users to understand how models make decisions by providing a closer look into the underlying factors driving predictions, encouraging greater confidence in model outcomes (Arrieta et al., 2020). This emphasis on explainable AI (XAI) practices not only improves model interpretability but also ensures that ecological decision-making processes are ethically sound (Xu et al., 2019). XAI has emerged as a critical paradigm in response to the increasing complexity and opacity of traditional AI models, which operate as black boxes, therefore aiming to provide explanations for AI decisions and predictions thanks to the alternative of white boxes or glass boxes (Franzoni, 2023; Das and Rad, 2020; Lucas, 2020; Rai, 2019). Through the enhancement of transparency and reliability, XAI allows users to understand and verify the reasoning behind AI-generated outcomes, not only improving accountability but also helping uncover biases (Chan, 2022), with the resulting transparency promoting fairness and equity in AI applications and encouraging trust in AI systems (Angerschmid et al., 2022; Shin, 2021). It is clear that incorporating ethical considerations – such as bias mitigation, fairness, and privacy protection – into AI development and use supports responsible AI frameworks that align with ecological conservation goals, ensuring the ethical use of these technologies in ecological research and decision-making processes (McGovern et al., 2022). Additionally, with the increasing focus on data privacy and ethics (Siau and Wang, 2020), regulations like the recently adopted EU AI Act (Laux et al., 2023) or the EU General Data Protection Regulation (GDPR) mandate the right to explanation for automated decisions (Scantamburlo et al., 2023), hence AI can ease regulatory compliance by providing interpretable explanations for AI outputs, helping identify weaknesses, errors, or areas for improvement in AI models. Lastly, in order to shed light on the reliability of predictions and guide more informed decision-making in the face of uncertainty, including metrics of model uncertainty helps improve credibility and trustworthiness as well as account for and manage uncertainty more effectively (Hoffman et al., 2018).

6.4. Causal inference

Models can sometimes yield correct predictions based on patterns in the data without capturing the true causal relationships, meaning that the model's predictive accuracy does not necessarily imply a deep understanding of the underlying causal mechanisms driving the outcomes (Pichler and Hartig, 2023). Therefore, caution should be exercised when interpreting the results of ML models as causal effects, due to features strongly used by the model not necessarily having a direct causal relationship with the outcome but could be instead correlated with other influential features in the dataset (Lederer et al., 2019; Genuer et al., 2010). To avoid this, removing some features can sometimes improve predictive performance, even if the features are causally related to the response variable (Arif and MacNeil, 2022). Regularization techniques were originally developed to address issues like collinearity and prevent overfitting (Hoerl and Kennard, 1970), however, while regularization can help mitigate collinearity problems (Chan et al., 2022), it does not guarantee the selection of the most causal features from a set of collinear variables (Zou and Hastie, 2005). Nonetheless, general ML and XAI can identify patterns and features in high-dimensional datasets that traditional statistical tools may struggle with, facilitating confirmatory analyses (Lucas, 2020; Buchelt et al., 2024; Pichler et al., 2019). Another option is to pre-select features consistent with causal inference, which improves ML's control over confounders (Zečević et al., 2021) and helps estimate causal effects of treatments (Chernozhukov et al., 2018). A third approach involves incorporating known physical or biological laws as constraints in the learning process of neural networks – as done with physics-informed neural networks (PINNs) – to adhere to established laws (Perry et al., 2022; Karniadakis et al., 2021). This is overall beneficial in terrestrial ecology, where adherence to ecological principles is needed to align general ML models with domain-specific knowledge (Wesselkamp et al., 2022). Ongoing research aims to develop ML methods specifically tailored for causal discovery (Pichler and Hartig, 2023). For instance, DL research has made progress in symbolic regression, which is a specialized form of genetic programming aiming to discover mathematical equations that accurately represent relationships within a dataset (Schmidt and Lipson, 2009). Genetic programming, more generally, is an evolutionary algorithm that simulates the process of natural selection to evolve computer programs or mathematical expressions for solving specific problems (Koza, 1992). In ecology, symbolic regression has been applied to model complex ecological relationships, such as species richness and spatial distributions, by generating free-form equations that capture the underlying data patterns. This method offers greater flexibility and interpretability compared to traditional statistical techniques, enabling the discovery of general models and principles applicable across various ecological systems (Cardoso et al., 2020).

6.5. Generative AI

A potentially powerful emerging tool is generative AI, which is capable of generating outputs that resemble existing data but do not constitute entirely new data, as it relies on learned patterns. Unlike traditional AI models that are primarily used for analysis, prediction, or classification, generative AI models are designed to produce novel outputs that can range from text and images to synthetic data for training other AI models (Nah et al., 2023). Producing synthetic datasets for training other AI models has been proposed as a potential approach in scenarios where real data is scarce or sensitive (Pichler and Hartig, 2023), suggesting that generative AI might be used to achieve performance close to human-like in specific ecological tasks, such as species recognition and distribution modelling. However, extreme caution is required when considering the use of AI-generated data for training models intended for prediction or inference. While its potential uses still remain unclear, ensuring that the generated content is high quality, free of errors or biases, and appropriate for scientific use is particularly

difficult and also raises ethical issues, as the generated outputs can significantly distort findings and lead to misleading conclusions (Haghighi et al., 2023). Generative AI models learn from patterns in the input data, which may already contain biases, inaccuracies, or incomplete information, thus making it challenging to verify that generated data is free of errors, as these models can inadvertently reproduce or even amplify such issues in the synthetic data they create (Feuerriegel et al., 2023). The inherent black-box nature of many generative models makes it additionally difficult to understand the exact process by which they generate content, making it hard to pinpoint and correct specific errors or biases (Franzoni, 2023). For instance, in species recognition tasks, if a generative AI model is trained on a dataset that underrepresents certain species or habitats, it might produce synthetic data that systematically exclude these species or misrepresent their features, leading to skewed results that might not fully capture the real-world variability or may even perpetuate existing gaps in knowledge.

Included within the ethical issues should also be environmental concerns, given the significant computational requirements associated with the increasingly widespread adoption of generative AI (and AI use across various fields in general), which can lead to substantial energy consumption and associated environmental impacts (Berthelot et al., 2024).

7. Conclusions

The future of AI and general ML applications in terrestrial ecology holds great promise, particularly in advancing real-time monitoring and near-instantaneous predictions, including seasonal forecasting, to respond more dynamically to environmental changes. These technologies can improve conservation and restoration strategies by providing more accurate, scalable, and timely data for ecosystem management, but this shift in how we approach, understand, and manage natural systems is not without challenges. The complexity, computational demands, and ethical concerns surrounding AI use – such as data biases and environmental costs, among others – necessitate a balanced approach. While AI offers unique tools for understanding ecological systems, its responsible implementation requires transparent and interpretable models that align with ecological principles. To fully harness the potential of ML, DL, and RL, continued innovation in AI methodologies tailored for ecological use, along with a concerted effort to invest in multi- and transdisciplinary collaborations, is critical to ensuring these technologies are not only effective but also sustainable and equitable in supporting ecosystem conservation and restoration, which are urgently needed amid such unprecedented environmental pressures.

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Declaration of generative AI writing

AI-assisted technologies have only been used to improve the readability of the manuscript.

The authors take full responsibility for the content of the published article.

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Cristina Cipriano: Writing – review & editing, Writing – original draft, Visualization, Methodology, Conceptualization. **Sergio Noce:** Writing – review & editing. **Simone Mereu:** Writing – review & editing. **Monia Santini:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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